

Adding storage to NEM 3.0 solar: winning back \$169k a year for a commercial site

California's Net Billing Tariff cut what exported solar earns. I built an hourly billing engine to measure how much that costs a commercial customer, and how much a right-sized battery recovers.

-\$56k/yr Lost when the same solar moves from NEM 2.0 to NEM 3.0	+\$169k/yr Recovered by adding a 500 kW / 2,000 kWh battery	84% Cut in July 4–9 pm peak demand, 759 kW to 124 kW	4.6 yr Simple payback on the battery alone, after the 30% ITC
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The question

Under NEM 2.0, a kWh exported at noon earned close to the retail rate. Under NEM 3.0 it earns the hourly Avoided Cost Calculator (ACC) value, which is lowest exactly when solar produces most. For this site the average export credit falls from **\$0.108** to **\$0.032/kWh**, a 70% drop.

The customer needs three answers: what NEM 3.0 costs them, whether storage fixes it, and what battery size pays back.

The site

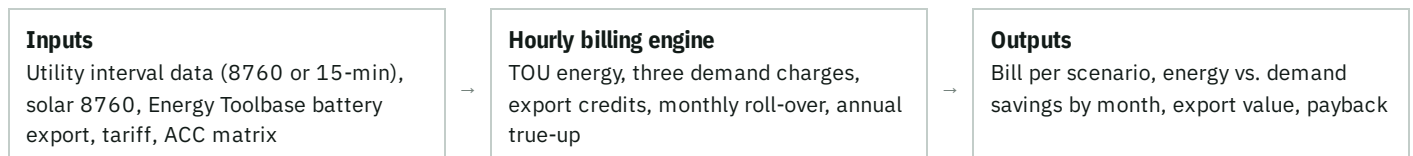
A synthetic Central Valley cold-storage warehouse: 3.6 GWh a year, 780 kW peak, an evening second shift. PV is 1,500 kWdc, covering 69% of annual energy. Rates follow the structure of PG&E B-19, with illustrative values.

Method

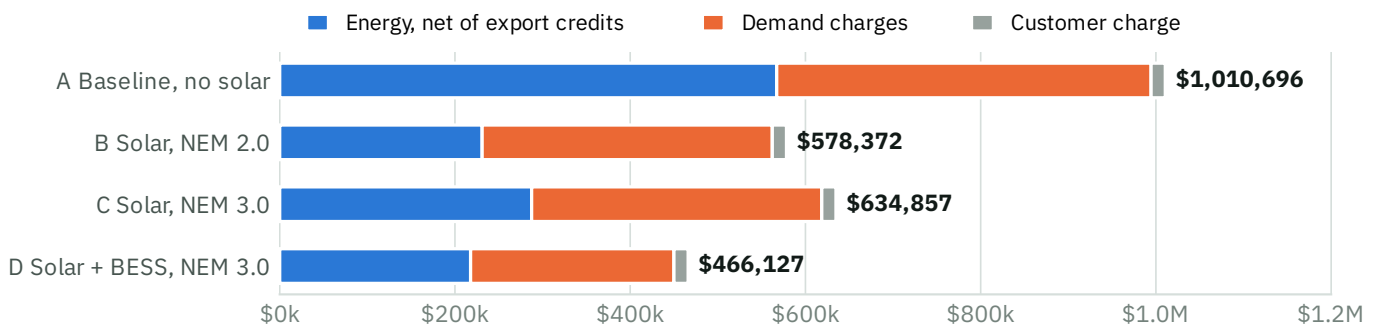
Every hour of the year is billed four ways on the same tariff:

1. **Baseline.** Load only. TOU energy, plus monthly maximum, peak and part-peak demand charges.
2. **Solar, NEM 2.0.** Exports credited at the retail TOU rate minus non-bypassable charges.
3. **Solar, NEM 3.0.** Exports credited at the ACC value for their month and hour.
4. **Solar + BESS, NEM 3.0.** Net load after battery dispatch, billed under NEM 3.0.

Credits roll month to month and true up annually. The battery charges only from surplus solar, then flattens the 4–9 pm window to the lowest import level its stored energy can hold. An Energy Toolbase dispatch export can replace this logic.

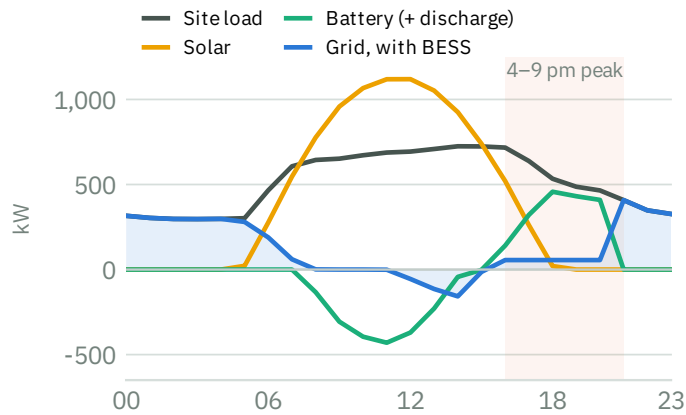


Annual bill by scenario



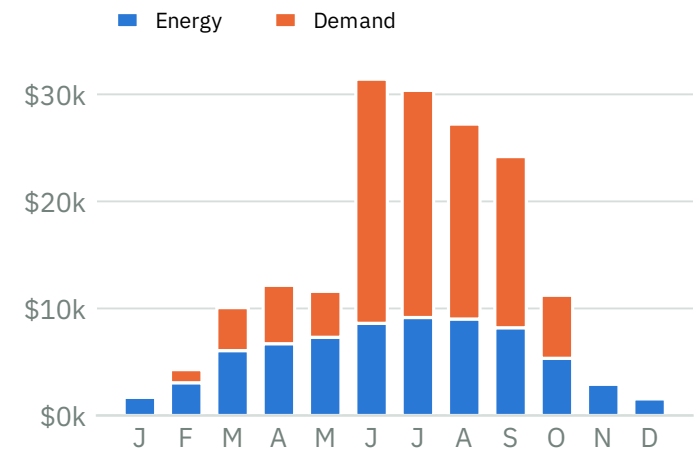
Solar alone cuts the bill by 37–43%. NEM 3.0 mainly erodes the export credit, shown in the energy segment. The battery adds its largest saving in demand charges.

How the battery earns: an average July weekday



Midday surplus that would export for about 3¢ charges the battery. The battery then holds grid import low through the 4–9 pm peak.

Monthly value of adding the battery to NEM 3.0 solar



Summer months carry most of the value through peak demand charges. In winter, little surplus solar reaches the battery.

Results

SCENARIO	ANNUAL BILL	ENERGY, NET	DEMAND	EXPORT MWH	SAVINGS	PAYBACK
A · Baseline, no solar	\$1,010,696	\$567,232	\$427,263	0	—	—
B · Solar, NEM 2.0	\$578,372	\$231,035	\$331,137	744	\$432,323	5.1 yr
C · Solar, NEM 3.0	\$634,857	\$287,520	\$331,137	744	\$375,838	5.9 yr
D · Solar + BESS, NEM 3.0	\$466,127	\$217,860	\$232,067	264	\$544,569	5.5 yr

Payback is simple, on year-one savings, after the 30% ITC, at illustrative installed costs of \$2.10/Wdc for PV and \$550/kWh for storage. It excludes degradation, rate escalation and O&M.

Findings

- **NEM 3.0 costs this site \$56k a year** on the same solar. Solar-only payback stretches from 5.1 to 5.9 years.
- **The battery recovers \$169k a year:** \$99k in demand charges and \$70k in energy. That is three times what NEM 3.0 took, so solar plus storage beats NEM 2.0 solar-only.
- **Exports fall from 744 to 264 MWh**, because the surplus is stored and used on site instead of sold at ACC values.
- **Winter is the weak season.** The battery discharges about 15 MWh a month from November to February, against about 49 MWh in summer. Winter peak demand barely moves.

Next steps on a real project

- Replace illustrative rates with the current tariff sheet and the customer's ACC vintage (export values lock for nine years).
- Use 15-minute interval data. Demand charges bill on 15-minute peaks, so hourly data understates them.
- Test winter grid charging in super-off-peak hours, and a demand cap for the non-peak maximum-demand charge.
- Run the sizing sweep in Energy Toolbase and feed the dispatch back through the same billing engine.

Tools: Python (pandas, NumPy), Energy Toolbase dispatch as input, and a browser version of the same engine. The two agree to the dollar.

Try it with your own interval data. Live tool: YOUR-USERNAME.github.io/solar-storage-workbench · Code: github.com/YOUR-USERNAME/solar-storage-workbench